

AI511/MM505 Linear Algebra with Applications

Lecture 1 – General Information, Systems of Linear Equations

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September 1, 2026

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Outline

General information

Systems of linear equations

- Motivation

- Definitions

- Two equations and two unknowns

- Row echelon form and Gauss-Jordan elimination

General information

Teachers and teaching assistant

Teacher of the theory part

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Teacher of the applications part

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Teaching assistant

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Two course codes

- AI511 is a 7.5 ECTS course that has two parts:
 - a 5 ECTS theory part (taught by Vaidotas Characiejus);
 - a 2.5 ECTS applications part (taught by Henry Kirveslahti).
- The 5 ECTS course MM505 coincides with the theory part of AI511.

Lectures and exercise sessions

Week	Lectures	Exercise sessions
36	Theory	No exercise sessions
37	Theory	Theory
38	Applications	Applications
39	Theory	Theory
40	Theory	Theory
41	Theory	Theory
42	Autumn break	
43	Theory	Theory
44	TBA later	Theory

- Week 38 is optional for the students of MM505.
- The complete schedule of the applications part will be announced by the applications part teacher.

Exercise sessions

- There will be weekly exercise sessions, except in week 36.
- The exercises will be made available about one week before each session.
- You are expected to prepare for each session by working through the assigned exercises and problems in advance.
- You are not expected to solve all exercises completely before the session. However, you should familiarise yourself with the problems, think about possible approaches, and make as much progress as you can.
- The teaching assistant will lead the sessions and help you work through the exercises.

Take-home assignment

- The exam of the theory part of AI511 will consist of a set of pen-and-paper problems to solve at home.
- The problems are expected to be posted in week 45 and you will have 72 hours to complete the assignment.
- This exam will have to be taken by the students of AI511 and by the students of MM505.
- For the students of AI511, there will be additional assignments in the applications part.

- Course announcements will be posted on itslearning.
- Lecture and exercise session material will be posted on <https://imada.sdu.dk/u/characiejus/linearalgebra2026/>
- The take-home assignment of the theory part will have to be downloaded and submitted on Digital Eksamen.

Textbook

Nathaniel Johnston (2021). *Introduction to linear and matrix algebra*.

Springer Cham

Animated series

Linear Algebra: An introduction to visualizing what matrices are really doing by 3Blue1Brown (freely available [here](#)).

- Systems of linear equations.
- Matrix operations, inverses, determinants.
- Vector spaces, basis, coordinates, linear independence.
- Linear transformations, eigenvalue problems, diagonalisation.
- Inner product and orthogonality.

Systems of linear equations

- Linear algebra is key mathematical tool for machine learning¹, statistics, optimisation², signal processing, computer vision, etc.
- The central problem of linear algebra is solving systems of linear equations.
- Why would we be interested in systems of linear equations?

¹AI512: Introduction to Machine Learning

²AI505: Optimisation

Systems of linear equations

Motivation

Example: optimising a quadratic function

- Consider the following function

$$f(x) = ax^2 + bx + c \quad (1)$$

for $x \in \mathbb{R}$, where $a, b, c \in \mathbb{R}$ are known constants.

- If $a \neq 0$, this function has a unique minimum or a unique maximum.
- How can we find x , where the value of the function is the smallest or the largest?

Example: optimising a quadratic function (cont.)

- The derivative³ is given by

$$f'(x) = 2ax + b$$

for $x \in \mathbb{R}$.

- By setting the derivative equal to 0, we obtain

$$2ax + b = 0. \tag{2}$$

- (2) is a *linear equation* that we need to solve to find a point where the function is at its unique minimum or at its unique maximum.
- Provided that $a \neq 0$, the solution of equation (2) is given by

$$x = -\frac{b}{2a}.$$

³Differentiation is covered in AI515: Calculus and statistics

Example: optimising a quadratic function (cont.)

- Consider another function

$$f(x_1, x_2) = a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2 + b_1x_1 + b_2x_2 + c,$$

for $x_1, x_2 \in \mathbb{R}$, where $a_{11}, a_{12}, a_{21}, a_{22}, b_1, b_2, c \in \mathbb{R}$ are known constants.

- Quadratic functions are behind probably the most commonly used loss functions (quadratic loss functions) in statistics and machine learning.
- We ask the same question: how can we find x_1 and x_2 , where the value of the function is the smallest or the largest?

Example: optimising a quadratic function (cont.)

- The partial derivatives⁴ are given by

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = 2a_{11}x_1 + (a_{12} + a_{21})x_2 + b_1$$

and

$$\frac{\partial f(x_1, x_2)}{\partial x_2} = (a_{12} + a_{21})x_1 + 2a_{22}x_2 + b_2.$$

- Setting both partial derivatives to 0, leads to a system of two linear equations with two unknowns

$$\begin{cases} 2a_{11}x_1 + (a_{12} + a_{21})x_2 + b_1 = 0; \\ (a_{12} + a_{21})x_1 + 2a_{22}x_2 + b_2 = 0. \end{cases} \quad (3)$$

⁴Partial derivatives are covered in AI515: Calculus and statistics

Example: optimising a quadratic function (cont.)

- When does a unique minimum or a unique maximum exist (corresponds to $a \neq 0$ in (1))?
- If either a minimum or a maximum exists, does the function have a minimum (corresponds to $a > 0$ in (1)) or a maximum (corresponds to $a < 0$ in (1))?
- If either a unique minimum or a unique maximum exists, how can we find it (i.e., how can we solve (3))?
- We need to know linear algebra to answer these questions.

Better notation

- Can we also use some better notation to make the problem clearer?
- We will learn that we can actually write

$$f(x) = x^T A x + b^T x + c,$$

where

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix},$$

and A^T denotes the transpose of the matrix A .

- The system of equations that we need to solve can then be compactly written as

$$(A + A^T)x + b = 0,$$

where $0 = \begin{pmatrix} 0 & 0 \end{pmatrix}^T$.

If $a_{12} = a_{21}$ (i.e., we will learn that such matrices are called symmetric), system of linear equations (3) can be expressed as

$$2Ax + b = 0 \tag{4}$$

which looks essentially identical to (2).

Wrap-up

- Systems of linear equations arise in problems that are not necessarily linear themselves, such as the optimization of a quadratic function.
- Optimising a quadratic function boils down to solving a system of linear equations.
- The matrix A determines how function f behaves (i.e., whether it has a unique minimum, a unique maximum, or neither).
- We also need to find the value of $x = (x_1 \ x_2)^T$ (if it exists) that satisfies (3) which means that we need to solve system of equations (3).
- It is a good idea to have a simpler problem in mind when we consider more complicated problems (compare (2) with (4)).

Systems of linear equations

Definitions

Linear equation

Definition

A *linear equation* in n variables $x_1, x_2, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b,$$

where $a_1, a_2, \dots, a_n$ and b are constants called the *coefficients* of the linear equation.

- The following equations are all linear:

$$x + 3y = 4, \quad \sqrt{3}x - y = \sqrt{5}, \quad \text{and} \quad \cos(1)x + \sin(1)y = 2.$$

- The following equations are all *not* linear:

$$\sqrt{x} + 3y = 4, \quad \cos x + \sin y = 0, \quad \text{and} \quad 4y + 2xz = 1.$$

Flat objects

- Geometrically, we can think of linear equations as representing lines and planes (and higher-dimensional flat shapes that we cannot quite picture).
- For example, the general equation of a line in $\mathbb{R}^2$ is

$$ax + by = c,$$

and the general equation of a plane in $\mathbb{R}^3$ is

$$ax + by + cz = d,$$

both of which are linear equations.

Section 2.1 of Johnston (2021)

System of linear equations

Definition

A system of linear equations (or a linear system) is a finite set of linear equations, each with the same variables $x_1, x_2, \dots, x_n$.

A general system of linear equations can be written as

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1; \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2; \\ \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{array} \right. \quad (5)$$

Section 2.1 of Johnston (2021)

Solutions and solution sets

Definition

A *solution* of a system of linear equations consists of n real numbers

$$s_1, s_2, \dots, s_n$$

such that the substitution $x_1 = s_1, \dots, x_n = s_n$ makes all m linear equations satisfied simultaneously.

Definition

The *solution set* of a system of linear equations is the set of all solutions of the system.

Definition

A system of linear equations that has at least one solution is called *consistent* and *inconsistent* otherwise.

Systems of linear equations

Two equations and two unknowns

Two equations and two unknowns

- Consider the following system of linear equations

$$\begin{cases} 5x + y = 3; \\ 2x - y = 4. \end{cases} \quad (6)$$

- One may solve this by isolating one variable in one of the equations and substituting it into the other equation to obtain $x = 1$ and $y = -2$.
- (6) is equivalent to

$$\begin{cases} y = -5x + 3; \\ y = 2x - 4. \end{cases} \quad (7)$$

- (7) describes two lines in the plane that intersect at the point $(1, -2)$ which is the solution of system of linear equations (6).

Existence and number of solutions

- System of equations (6) has exactly one solution.
- There might be no solution:

$$\begin{cases} 2x + y = 3; \\ 2x + y = 5. \end{cases} \quad (8)$$

- There might be infinitely many solutions:

$$\begin{cases} 2x + y = 3; \\ 4x + 2y = 6. \end{cases} \quad (9)$$

- What happens geometrically when there is no solution and when there are infinitely many solutions?
- There are only three possibilities: a system of linear equations can have none, one, or infinitely many solutions.

Systems of linear equations

Row echelon form and Gauss-Jordan
elimination

Triangular shape and back substitution

- If the linear system has a “triangular” form, then solving it is fairly intuitive.

- For example,

$$\begin{cases} x + 3y - 2z = 5; \\ 2y - 6z = 4; \\ 3z = 6. \end{cases} \quad (10)$$

- The solution is $z = 2$, $y = 8$, and $x = -15$.
- The procedure that we used to solve the example is called *back substitution*, and it worked because of the triangular shape of the linear system.
- Our goal is to put *every* system of equations into this triangular form.

Transformations of systems of linear equations

- Consider another system of linear equations

$$\begin{cases} x + 3y - 2z = 5; \\ x + 5y - 8z = 9; \\ 2x + 4y + 5z = 12. \end{cases} \quad (11)$$

- Can we somehow transform (11) into the triangular shape?
- The following three operations do not affect the set of solutions (think why this is the case):
 - (a) multiplying an equation by a non-zero scalar;
 - (b) swapping the order of equations;
 - (c) adding a scalar multiple of one equation to another.
- We can actually transform (11) into (10) without affecting the set of solutions (example on the blackboard).